

When Does Non-Myopic Bayesian Experimental Design Work with Language Models?

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Abstract

Sequential Bayesian experimental design (BED) offers a principled way for language-model agents to ask questions, update beliefs, and plan future information gathering. Yet deeper planning can optimize errors in an LLM-derived probabilistic model rather than useful information. We audit this failure across an exact constrained-location test, semantic 20 Questions, and two external interactive benchmarks: Paprika troubleshooting and MediQ clinical reasoning. Depth-two planning succeeds in the exact test (final RMSE 0.153 versus 0.561 for greedy EIG), while one-step EIG substantially beats prompting in 20 Questions. The incremental depth effect does not survive paired evaluation, and depth three is worse. Paprika then exposes an objective and endpoint mismatch; MediQ passes a strict simulator audit but fails likelihood calibration, with only 8 of 30 held-out questions answerable from its static records. A preregistered iCRAFT-MD diagnosis-profile repair then fails at bounded latent-support construction. These results identify two independent prerequisites for non-myopic LLM BED: a deployment-matched generative model whose latent state is sufficient for responses, and a task with a verified non-greedy information structure. We formulate a validation ladder that tests these prerequisites before scaling horizon or rollouts. The main lesson is not that lookahead is ineffective, but that planning depth is downstream of model validity and a measurable greedy gap.

1 Introduction

Interactive agents must often acquire information before committing to an answer or action. A clinical agent asks about symptoms; a support agent diagnoses a fault; a guessing agent narrows a large semantic space. Direct prompting is an unreliable solution: MediQ reports that asking models to interact can reduce accuracy relative to answering from the initial context [Li et al., 2024]. Bayesian experimental design instead selects a query by its expected information gain (EIG) about a target variable. BED-LLM makes this practical in semantic spaces by using an LLM to propose hypotheses and provide response likelihoods [Choudhury et al., 2026].

The natural next step is non-myopic BED: score a query by both its immediate EIG and the information available to a future adaptive query. Recent methods simulate conversation trees for this purpose [Hu et al., 2024, Arnould et al., 2026]. In principle this can value a prerequisite question with little immediate gain. In practice every additional branch invokes the same uncertain model, changes the belief on which later calls condition, and maximizes over more noisy continuation scores. A larger tree can therefore make a confidently worse decision.

We investigate this distinction using a validation-first sequence of experiments. An exact constrained-location environment verifies that our depth-two recursion can beat greedy EIG when both the model and delayed-information trap are exact. Semantic 20 Questions verifies one-step transfer but does not establish an incremental horizon benefit. Natural location experiments show non-monotonic depth under LLM-generated strategies. Finally, Paprika and MediQ expose different failures on external benchmarks: task utility and endpoint attribution in Paprika, and latent-state sufficiency plus record sparsity in MediQ.

Our contributions are:

1. a cross-environment audit that separates correctness of the planning recursion from validity of the probabilistic interaction model;
2. paired evidence that one-step semantic BED can help while increasing horizon need not, alongside an exact positive control where lookahead does help;
3. two external-benchmark autopsies with invalid outcomes quarantined rather than repurposed as method evidence; and
4. a concrete validation contract for future non-myopic LLM BED: target and latent sufficiency, calibrated likelihoods, branch/deployment equivalence, a verified greedy gap, and ranking fidelity before policy evaluation.

2 The Probabilistic Contract

Let $h_t = ((x_1, y_1), \dots, (x_t, y_t))$ be the interaction history and θ the finite target decoded at the endpoint. A generative state z may contain additional information needed to predict responses, with $\theta = g(z)$. The relevant model is

$$p_t(\theta, z)p(y | z, x, h_t), \quad p(y | \theta, x, h_t) = \sum_z p(y | z, x, h_t)p_t(z | \theta).$$

When θ itself is sufficient, as for a hidden animal answering a property question, z can be omitted. When θ is merely a decision label, such as “treat hypoperfusion first,” it does not determine a patient’s glucose or history and cannot serve as the response-generating state.

For a coherent model, one-step EIG is

$$I_t(x) = \sum_{\theta, y} p_t(\theta)p(y | \theta, x, h_t) \log \frac{p(y | \theta, x, h_t)}{\sum_{\theta'} p_t(\theta')p(y | \theta', x, h_t)}.$$

Our exact depth-two score is

$$Q_2(x) = I_t(x) + \mathbb{E}_{y|x, h_t} \left[\max_{x' \in \mathcal{X}(h_t, x, y)} I_{t+1}(x') \right].$$

It branches over every positive-mass response, performs a Bayesian update on the fixed target support, generates branch-specific follow-ups, and adds the best second-step EIG. This is mathematically appropriate for the model supplied to it. It cannot determine whether that model describes deployment.

Endpoint target $\theta \rightarrow$ sufficient state $z \rightarrow$ calibrated $p(y | z, x, h) \rightarrow$ matched simulated/deployed update $\rightarrow$ verified greedy gap $\rightarrow$ rankable depth increment $\rightarrow$ endpoint comparison

Figure 1: The validation chain for non-myopic LLM BED. Failure of any upstream link invalidates evidence from downstream policy performance.

Two requirements are therefore distinct. *Model validity* asks whether the target, latent state, likelihood, update, and simulated branches form one process that matches deployment. *Planning value* asks whether the task contains complementarity or gating such that a lower-immediate-EIG query enables a more informative continuation. If information has adaptive diminishing returns, greedy policies can already be competitive [Golovin and Krause, 2011]. Depth needs a demonstrated gap, not merely a multi-turn interface.

3 Audit Design and Evidence Rules

We aggregate only preregistered or reconstructable paired evidence. Exact location tests use the known source prior and signal likelihood. The animals audit reconstructs every 20-round binary trial and reports deterministic bootstrap intervals. Paprika and MediQ use hash-pinned official releases and strict transcript audits. Failed parsing, unsupported simulator facts, and endpoint contradictions fail closed.

Most importantly, we distinguish *policy evidence* from *diagnostic evidence*. A Paprika run whose customer reports the private correct remedy as failing cannot compare policies, even if aggregate resolution numbers exist. A MediQ run with an invalid likelihood cannot establish EIG value, even if its queries are fluent. Such runs may localize a broken link in Figure 1; they do not support an accuracy claim.

Setting	Generative model	Greedy gap	Key result	Evidence status
Exact constrained location	Exact	Explicit delayed trap	RMSE: d2 .153, greedy .561	Positive control
Animals 20Q	Concrete target, binary replies	Unverified	d1 AUC .743, think-naive .465	One-step positive
Natural location	Exact signal, LLM strategies	Weak/variable	d1/d3/d5 non-monotonic	Horizon negative
Paprika	Insufficient state/task utility	Assumed, not enforced	Endpoint contradiction	Diagnostic only
MediQ iMEDQA	A–D often not generative	Channel too sparse	8/30 replies available	Gate failure

Table 1: The same recursion behaves differently as the probabilistic and task contracts weaken. “d1” and “d2” denote scoring horizon.

4 Results Across Environments

Exact lookahead can work. We first constructed a branch-decoy source prior with a local-bump observation model and movement-constrained queries. The first query can reveal which region to enter, while movement limits make that choice affect later measurements. Over 120 paired trials, exact depth two reduced final RMSE from 0.5606 for greedy EIG to 0.1529, a mean paired difference of -0.4077 . This is a harness result, not an external-benchmark claim. It shows that the implementation and objective can exploit a genuine delayed-information structure when model error is absent.

One-step semantic BED transfers; depth does not yet. On animals 20 Questions, depth-one EIG achieved accuracy AUC 0.743 versus 0.465 for a thinking naive policy and 0.458 for non-thinking naive over 40 paired trials. The AUC gain over thinking naive was 0.278 with 95% bootstrap interval $[0.191, 0.365]$. This is the stable positive BED result.

The apparent aggregate depth-two AUC was 0.770, but the full depth-one and depth-two streams were not paired. On the only exact ten-trial seed block, depth two minus depth one was -0.040 with interval $[-0.105, 0.025]$ and 3/1/6 wins/ties/losses. Depth three was worse than depth two across 40 paired trials: -0.126 $[-0.224, -0.035]$. Thus these data support semantic EIG, not non-myopic improvement.

Natural location experiments reinforce this distinction. In a 30-trial movement-constrained study using exact physical likelihoods but LLM-generated strategies, final RMSE was 0.412 for greedy EIG and 0.929, 1.127, and 0.542 for strategy horizons 1, 3, and 5. Without the movement constraint all horizons were effectively flat. More planning did not yield monotonic behavior, and the naive policy remained competitive.

Paprika fails before a horizon claim. Paprika troubleshooting mixes information-gathering questions with corrective actions that change and may terminate the world [Tajwar et al., 2025]. Pure EIG values discrimination among cause/remedy strings, not the probability of resolution minus action cost. A remedy that resolves under every hypothesis can have zero EIG, while a diagnostic that identifies the cause but cannot resolve the task can score highly.

The implementation also used fixed support and synthetic outcome labels in lookahead while deployment refreshed support, observed free text, and could terminate. A ten-task full-depth diagnostic used 70,004 requests versus 3,365 for one-step EIG and resolved 2 versus 3 tasks. Those counts are not policy evidence: later manual audit found a customer response claiming that the exact private remedy had failed. A separate 50-task arbitration evaluation was also quarantined by endpoint review. The defensible result is the failure mechanism, not which arm won an invalid simulator contest.

MediQ validates the interface but rejects the joint model. MediQ’s official FactSelect patient returns only atomic record facts that answer the question and otherwise abstains [Li et al., 2024]. After several failed closed repairs, the final five-case environment gate achieved clean canonical Yes/No/unavailable questions, verbatim grounding, relevance, and semantic deduplication on every selected interaction. This isolates the remaining issue to probability modeling.

The first scorer treated each A–D option as a complete patient world. Across ten selected turns, predicted EIG averaged 0.140 nats but realized true-label log-probability gain averaged -0.290 nats; the realized outcome was less likely under the true label than under the predictive mixture on 6/10 turns. A repair made record missingness label-independent and scored only Yes versus No under each option. Missingness became exactly posterior-neutral, yet available outcomes still had mean true-label log gain -0.118 nats.

A final data-estimation construction elicited the response marginal and hypothetical A–D posteriors, then projected their joint table to exact local marginals. Before scoring it, a frozen held-out bank required at least 15 available responses. All 30 questions were canonical, unique, grounded, and relevant, but only 8 were answerable; 22 facts were absent from the static records. The scorer was therefore not run. At observed answerability $a = 8/30$, an unavailable-neutral binary channel has the loose ceiling

$$I(\theta; Y \mid x) \leq a \log 2 = 0.1848 \text{ nats per query,}$$

before accounting for diagnostic ambiguity. There is little signal on which a depth-two advantage could operate.

We then ran a fresh, preregistered iCRAFT-MD diagnosis-only profile gate. The official scorer uses a four-way diagnosis index and the records are denser, so we assigned three concrete counterfactual patient profiles to each diagnosis, held record availability label-independent, and froze support before any branch. The one-case micro-smoke passed: it produced all 12 profiles, two candidates, and canonical profile likelihood tables for \$0.0027. On the held-out

calibration partition, however, the non-thinking 26B scaffold could not produce the required three valid profiles per diagnosis within the bounded repair budget. This fails the upstream profile-support gate before likelihood, branch-equivalence, greedy-gap, or policy evaluation. We made no prompt repair or rerun under the frozen protocol.

5 Why Horizon Magnifies Model Error

Let an estimated continuation score be $\widehat{V}_j = V_j + \epsilon_j$ for branch candidate j . Depth two uses $\max_j \widehat{V}_j$, so even zero-mean errors produce positive selection bias. Increasing branch width or horizon adds more opportunities for an optimistic likelihood, posterior, or proposal to win. Root likelihood error also changes branch mass and the posterior context passed to every downstream call. More Monte Carlo rollouts reduce variance conditional on this model; they do not reduce its bias. Bayesian design is particularly exposed to misspecification because the same model both predicts data and decides which data to collect [Catanach and Das, 2023].

BED-LLM reaches strong one-step results under cleaner conditions: a concrete entity or user profile, a small response space, a prior-likelihood construction, and diverse hypotheses retained with uniform mass after history filtering [Choudhury et al., 2026]. It explicitly reports raw in-context beliefs as overconfident and finds its data-estimation ablation weaker than prior-likelihood BED. Our MediQ belief similarly reached mean top probability 0.867 after round one at only 50% accuracy.

UoT reports improvements with simulation depth, but operates on explicit possibility sets and repeatedly partitions the same support; its diagnosis tasks contain only 5 or 15 diseases [Hu et al., 2024]. CA-BED uses finite mutually exclusive hypotheses, soft LLM likelihoods, binary questions, and depth two on structured entity deduction [Arnould et al., 2026]. It improves over direct prompting, but does not report a causal depth-one versus depth-two ablation; its categorical-answer extension hurts the harder Detective benchmark. These papers show that structured planning can help, not that horizon is robust to an insufficient latent or incompatible answer categories.

Deep adaptive design and RL-BED optimize total sequential utility through amortized policies or reinforcement learning [Foster et al., 2021, Blau et al., 2022]. Both require trajectories from a meaningful simulator. Training on the current MediQ surrogate would amortize its annotation and calibration artifacts rather than recover a missing patient state.

6 A Validation-First Path Forward

Future depth studies should cross the following gates in order:

1. **Target and state:** the endpoint must decode θ , while a latent z must contain everything required to generate responses.
2. **Prior:** held-out log loss and Brier score must beat uniform, with no high-confidence wrong collapse.
3. **Likelihood:** available observations must improve mean true-target log probability; missingness must be neutral unless its semantics are genuinely target-dependent.
4. **Branch equivalence:** simulated and deployed histories must yield the same support, probabilities, and stopping behavior.
5. **Structural gap:** an exact or high-fidelity oracle must show depth two beating greedy under the benchmark’s native action and budget rules.
6. **Ranking fidelity:** estimated incremental lookahead value must rank oracle incremental value under shared candidates and common random numbers.
7. **Policy test:** only then compare depth one and two on paired cases, with identical root candidates, simulator seeds, and stopping rules.

Within MediQ, the official iCRAFT-MD split offered a fresh diagnosis-only target: all 140 targets are diagnoses and records have a mean 14.82 atomic facts (median 14, minimum 9), compared with 11.20 (median 10, minimum 2) in usable iMEDQA. We preregistered and tested the natural repair—several concrete counterfactual patient profiles z per diagnosis, profile-conditioned replies, label-independent availability, and EIG about $\theta = g(z)$ —but the model did not reliably construct the fixed support on held-out cases. This is an upstream model-validity failure, not evidence that a later policy would fail. It nevertheless closes this un-tuned path: we do not promote iCRAFT as an unevaluated escape hatch.

7 Limitations and Conclusion

This is a validation study, not a positive external-benchmark claim for non-myopic LLM BED. The exact location gap is deliberately constructed and serves only as a correctness control. Animals establishes one-step transfer, but its full depth-one and depth-two streams are only partially paired and have weaker code provenance than the newer audits. Natural location results use one model family and environment. Paprika policy counts are endpoint-invalid, and MediQ stops before a calibrated policy comparison. We do not claim that non-myopic BED cannot work; we report that the preregistered iCRAFT profile construction did not pass its first held-out support gate.

The audits also cannot cleanly separate model capacity from representation: a larger model may estimate a better likelihood, but cannot make an exam-treatment label a sufficient patient state or reveal facts absent from a static record. OpenRouter provider nondeterminism, shared model-family errors, finite generated supports, and post-hoc synthesis across heterogeneous experiments remain limitations.

The evidence supports a narrower conclusion. Non-myopic recursion works when a coherent simulator and delayed-information gap are present. One-step semantic EIG can improve information gathering. But depth is not a substitute for a valid joint model, deployment-matched branches, or a task in which greedy information gain is genuinely suboptimal. These are empirical gates, not implementation details. Testing them before scaling horizon is both cheaper and scientifically stronger than explaining another non-monotonic depth sweep after the fact.

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